



Mata Kuliah : Pemrograman Web Lanjut (PWL)
Program Studi : D4 – Teknik Informatika
Semester : 4

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Jobsheet Ke- : 5

Laporan Jobsheet 05 – 03

Link Repository : <https://github.com/athfizh/Pemrograman-Web-Lanjut>

Praktikum

Langkah	Jawaban/Deskripsi
1	Membuat Model & Migration Category <pre>PS C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app> php artisan make:model Category -m</pre> INFO Model [C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app\app\Models\Category.php] created successfully. INFO Migration [C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app\database\migrations\2026_03_30_173507_create_categories_table.php] created successfully.
2	Mendesain Tabel Categories <pre>filament-app > database > migrations > 2026_03_30_173507_create_categories_table.php</pre> <pre>1 <?php 2 3 use Illuminate\Database\Migrations\Migration; 4 use Illuminate\Database\Schema\Blueprint; 5 use Illuminate\Support\Facades\Schema; 6 7 return new class extends Migration { 8 /** 9 * Run the migrations. 10 */ 11 public function up(): void 12 { 13 Schema::create('categories', function (Blueprint \$table) { 14 \$table->id(); 15 \$table->string('name'); 16 \$table->string('slug'); 17 \$table->timestamps(); 18 }); 19 } 20 21 /** 22 * Reverse the migrations. 23 */ 24 public function down(): void</pre>



3	Mengatur Model Category <pre>filament-app > app > Models > Category.php > ... 1 <?php 2 3 namespace App\Models; 4 5 use Illuminate\Database\Eloquent\Model; 6 7 0 references 0 implementations 8 class Category extends Model 9 { 10 // 11 0 references 12 protected \$fillable = [13 "name", 14 "slug", 15]; 16 }</pre>
4	Membuat Model & Migration Post <pre>PS C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app> php artisan make:model Post -m INFO Model [C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app\app\Models\Post.php] created successfully. INFO Migration [C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app\database\Migrations\2026_03_30_173953_create_posts_table.php] created successfully.</pre> Edit migration post menjadi:



```
filament-app > database > migrations > 2026_03_30_173953_create_posts_table.php
3 use Illuminate\Database\Migrations\Migration;
6
7 return new class extends Migration {
8     /**
9      * Run the migrations.
10     */
11     public function up(): void
12     {
13         Schema::create('posts', function (Blueprint $table) {
14             $table->id();
15             $table->string('title');
16             $table->string('slug');
17             $table->integer('category_id');
18             $table->string('color')->nullable();
19             $table->string('image')->nullable();
20             $table->text('body')->nullable();
21             $table->json('tags')->nullable();
22             $table->boolean('published')->default(false);
23             $table->date('published_at')->nullable();
24             $table->timestamps();
25         });
26     }
27 }
28 /**
```

```
PS C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app> php artisan migrate

INFO Running migrations.

2026_03_30_173953_create_posts_table ..... 32.54ms DONE
```

5

Mengatur Model Post

```
filament-app > app > Models > Post.php > ...
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 0 references | 0 implementations
8 class Post extends Model
9 {
10     //
11     0 references
12     protected $fillable = [
13         "title",
14         "slug",
15         "category_id",
16         "color",
17         "image",
18         "body",
19         "tags",
20         "published",
21         "published_at",
22     ];
23 }
```

Menggunakan Casting Laravel



```
filament-app > app > Models > Post.php > Post
0 references | 0 implementations
7  class Post extends Model
8  {
9      //
10     0 references
11     protected $fillable = [
12         "title",
13         "slug",
14         "category_id",
15         "color",
16         "image",
17         "body",
18         "tags",
19         "published",
20         "published_at",
21     ];
22     0 references
23     protected $casts = [
24         "tags" => "array",
25         "published" => "boolean",
26         "published_at" => "date",
27     ];
28 }
```

Menambahkan Relasi (pada model/Post.php)

```
filament-app > app > Models > Post.php > ...
7  class Post extends Model
10     protected $fillable = [
14         "color",
15         "image",
16         "body",
17         "tags",
18         "published",
19         "published_at",
20     ];
21     0 references
22     protected $casts = [
23         "tags" => "array",
24         "published" => "boolean",
25         "published_at" => "date",
26     ];
27     0 references | 0 overrides
28     public function category()
29     {
30         return $this->belongsTo(Category::class);
31     }
32 }
```

Membuat Resource Category di Filament



```
PS C:\laragon\www\Pemrograman-Web-Lanjut\jobsheet5\filament-app> php artisan make:filament-resource Category

The "title attribute" is used to label each record in the UI.

You can leave this blank if records do not have a title.

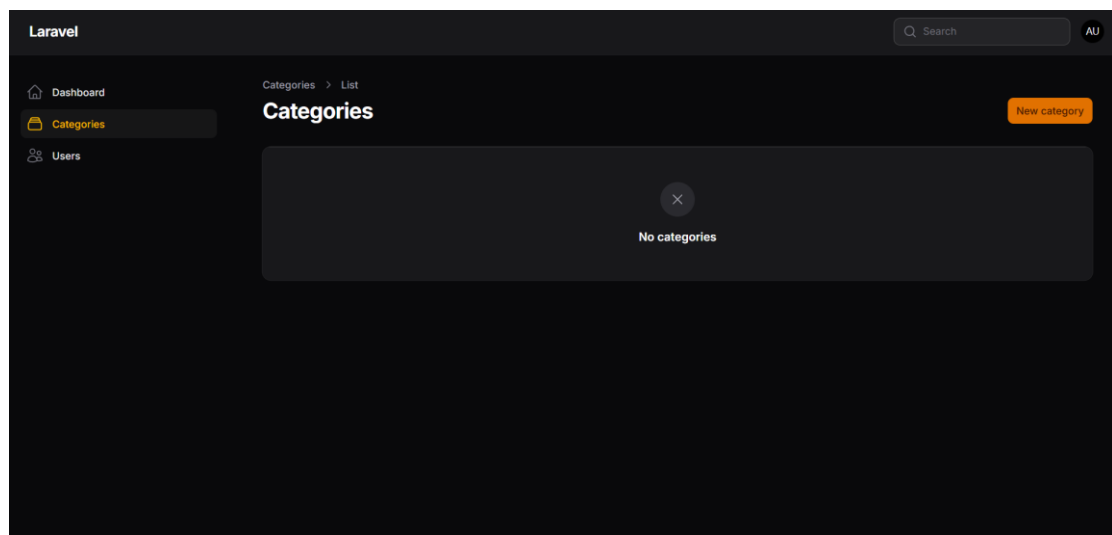
What is the title attribute for this model?
> name

Would you like to generate a read-only view page for the resource? (yes/no) [no]
> no

Should the configuration be generated from the current database columns? (yes/no) [no]
> no

[INFO] Filament resource [App\Filament\Resources\Categories\CategoryResource] created successfully.
```

Hasil sidebar categories:



Edit Form Category

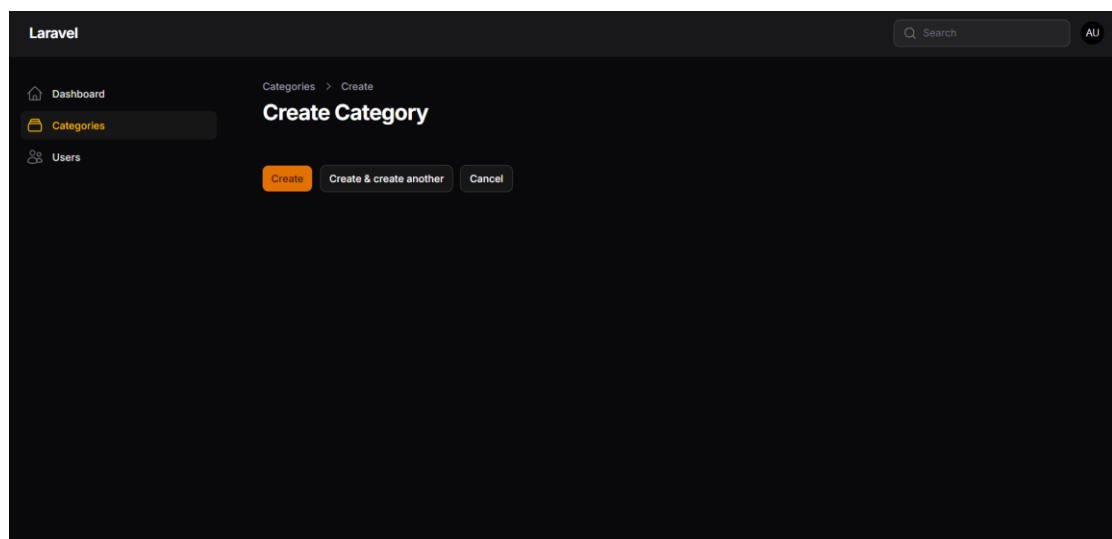


```
filament-app > app > Filament > Resources > Categories > Schemas > CategoryForm.php > CategoryForm > configure()

4
5 use Filament\Schemas\Schema;
6 use Filament\Forms\Components\TextInput;
7
8 2 references | 0 implementations
9 class CategoryForm
10 {
11     1 reference | 0 overrides
12     public static function configure(Schema $schema): Schema
13     {
14         return $schema
15             ->components([
16                 //
17                 TextInput::make('name')
18                     ->required(),
19                 TextInput::make('slug')
20                     ->required(),
21             ]);
22     }
23 }
```

Hasil:

Sebelum



Sesudah



Edit Table Category

```
filament-app > app > Filament > Resources > Categories > Tables > CategoriesTable.php > CategoriesTable > configure()
1 use Filament\Actions\EditAction;
2 use Filament\Tables\Table;
3 use Filament\Tables\Columns\TextColumn;
4
5 2 references | 0 implementations
6 class CategoriesTable
7 {
8     1 reference | 0 overrides
9     public static function configure(Table $table): Table
10    {
11        return $table
12            ->columns([
13                //
14                TextColumn::make('name'),
15                TextColumn::make('slug'),
16            ]) ->filters([
17                //
18            ])
19            ->recordActions([
20                EditAction::make(),
21            ])
22    }
23 }
```

Hasil:



The top screenshot shows the 'Create Category' form in a Laravel admin interface. The form has two input fields: 'Name' and 'Slug', both containing the value 'laravel'. Below the fields are three buttons: 'Create', 'Create & create another', and 'Cancel'. The sidebar on the left shows 'Dashboard', 'Categories', and 'Users' links. The top navigation bar shows a search bar and a user profile icon labeled 'AU'.

The bottom screenshot shows the 'Categories' list view in the same Laravel admin interface. It displays a table with one row: 'laravel' with slug 'laravel'. There is an 'Edit' button next to the row. The sidebar and top navigation bar are the same as in the top screenshot.

Pertanyaan

1. Mengapa kita perlu \$fillable?

Fungsi utama \$fillable adalah sebagai fitur keamanan (mencegah *Mass Assignment Vulnerability*). Fitur ini mendefinisikan secara eksplisit kolom database mana saja yang diizinkan untuk diisi atau diperbarui secara langsung melalui form/kode, sehingga kolom sensitif seperti id atau atribut admin tidak bisa dimanipulasi oleh *user* dari luar.

2. Apa fungsi \$casts pada Laravel?

Berfungsi sebagai *translator* (pengubah tipe data) otomatis antara database dan aplikasi. Contohnya, kolom tags yang disimpan sebagai *teks JSON* di database akan otomatis diubah



menjadi tipe data *Array* saat ditarik ke aplikasi, sehingga developer bisa langsung mengolahnya tanpa perlu mem- *parsing* secara manual.

3. Apa perbedaan integer biasa dengan foreign key?

- **Integer biasa:** Hanya menyimpan nilai angka bebas tanpa batasan logika.
- **Foreign Key:** Sebuah *integer* khusus yang diikat (berelasi) secara ketat dengan ID tabel lain. *Foreign key* menjaga integritas data; artinya, sistem akan menolak jika kita memasukkan ID kategori yang tidak pernah ada di tabel asal.

4. Bagaimana jika category dihapus tetapi masih ada post?

Tergantung aturan pada *foreign key*-nya.

- Jika diatur secara *strict* (bawaan database), maka sistem akan menolak penghapusan dan memunculkan **Error**, karena akan menyebabkan data Post memiliki *category_id* yatim piatu (*orphan*).
- Jika diatur dengan *cascadeOnDelete()*, maka semua tabel Post yang terhubung dengan Category tersebut akan ikut terhapus secara otomatis.

Tugas Praktikum

1. Tambahkan minimal 3 kategori berbeda



filament-app > database > seeders > CategorySeeder.php > ...

```
1  <?php
2
3  namespace Database\Seeders;
4
5  use App\Models\Category;
6  use Illuminate\Database\Seeder;
7
8  1 reference | 0 implementations
9  class CategorySeeder extends Seeder
10 {
11     /**
12      * Run the database seeds.
13      */
14     0 references | 0 overrides
15     public function run(): void
16     {
17         $categories = [
18             [
19                 'name' => 'Teknologi',
20                 'slug' => 'teknologi',
21             ],
22             [
23                 'name' => 'olahraga',
24                 'slug' => 'olahraga',
25             ],
26             [
27                 'name' => 'Kesehatan',
28                 'slug' => 'kesehatan',
29             ],
30             [
31                 'name' => 'Pendidikan',
32                 'slug' => 'pendidikan',
33             ],
34         ];
35
36         foreach ($categories as $category) {
37             Category::firstOrCreate(
38                 ['slug' => $category['slug']],
39                 $category
40             );
41         }
42     }
43 }
```

2. Tambahkan validasi slug harus unik



```
filament-app > app > Filament > Resources > Categories > Schemas > CategoryForm.php > CategoryForm > configure()
1  <?php
2
3  namespace App\Filament\Resources\Categories\Schemas;
4
5  use App\Models\Category;
6  use Filament\Schemas\Schema;
7  use Filament\Forms\Components\TextInput;
8
9  2 references | 0 implementations
10 class CategoryForm
11 {
12     1 reference | 0 overrides
13     public static function configure(Schema $schema): Schema
14     {
15         $record = $schema->getLivewire()->getRecord();
16
17         return $schema
18             ->components([
19                 TextInput::make('name')
20                     ->required()
21                     ->maxLength(255),
22                 TextInput::make('slug')
23                     ->required()
24                     ->maxLength(255)
25                     ->unique(
26                         table: Category::class,
27                         column: 'slug',
28                         ignorable: $record,
29                     )
30                 ->helperText('Slug harus unik, contoh: teknologi, olahraga'),
31             ]);
32 }
```

3. Ubah category_id menjadi foreign key



```
filament-app > database > migrations > 2026_03_30_173953_create_posts_table.php
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration {
8      /**
9       * Run the migrations.
10      */
11     public function up(): void
12     {
13         Schema::create('posts', function (Blueprint $table) {
14             $table->id();
15             $table->string('title');
16             $table->string('slug');
17             $table->foreignId('category_id')->constrained('categories')->cascadeOnDelete();
18             $table->string('color')->nullable();
19             $table->string('image')->nullable();
20             $table->text('body')->nullable();
21             $table->json('tags')->nullable();
22             $table->boolean('published')->default(false);
23             $table->date('published_at')->nullable();
24             $table->timestamps();
25         });
26     }
27
28     /**
29      * Reverse the migrations.
30      */
31     public function down(): void
32     {
33         Schema::dropIfExists('posts');
34     }
35 };
```

Kendala : -